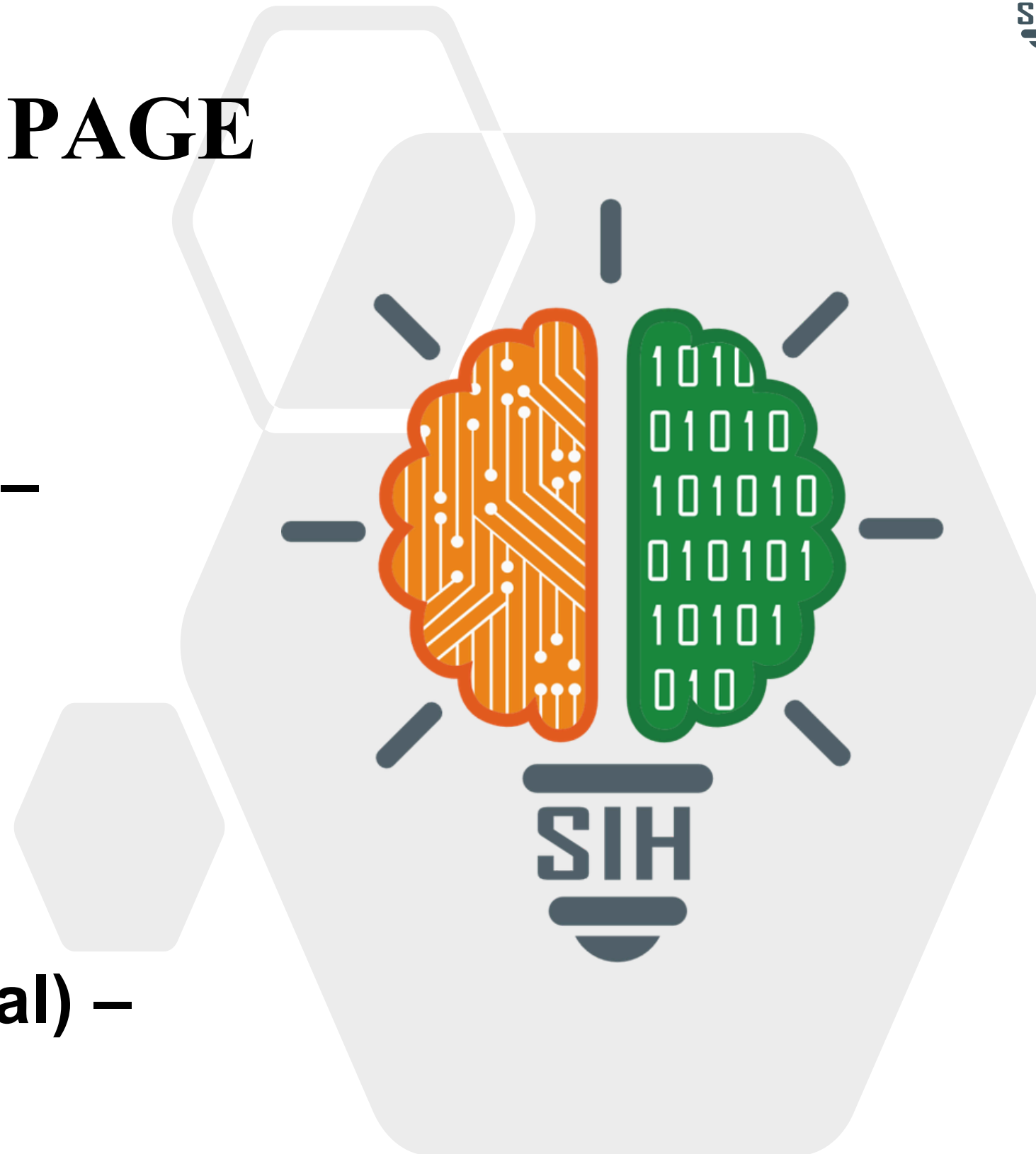


TITLE PAGE

- Problem Statement ID –
- Problem Statement Title- AURA –
Democratizing AQI Readings
- Theme- Open Innovation
- PS Category- Software
- Team ID-
- Team Name (Registered on portal) –
The Code Alchemists



The Problem



Pollution-linked fatalities in India exceed 1.6 million every year.



Vast regions remain unmonitored due to a critical lack of sensing infrastructure.



Smart relays prevent energy waste by only activating purifiers when AQI thresholds are breached.

Proposed Solution

AURA is a three layer architecture that **decentralizes AQI monitoring**, uses **blockchain to ensure data integrity** and provide an **IOT-based automated remediation** through air quality control units.

LAYER 1: EDGE IOT & ACTUATION

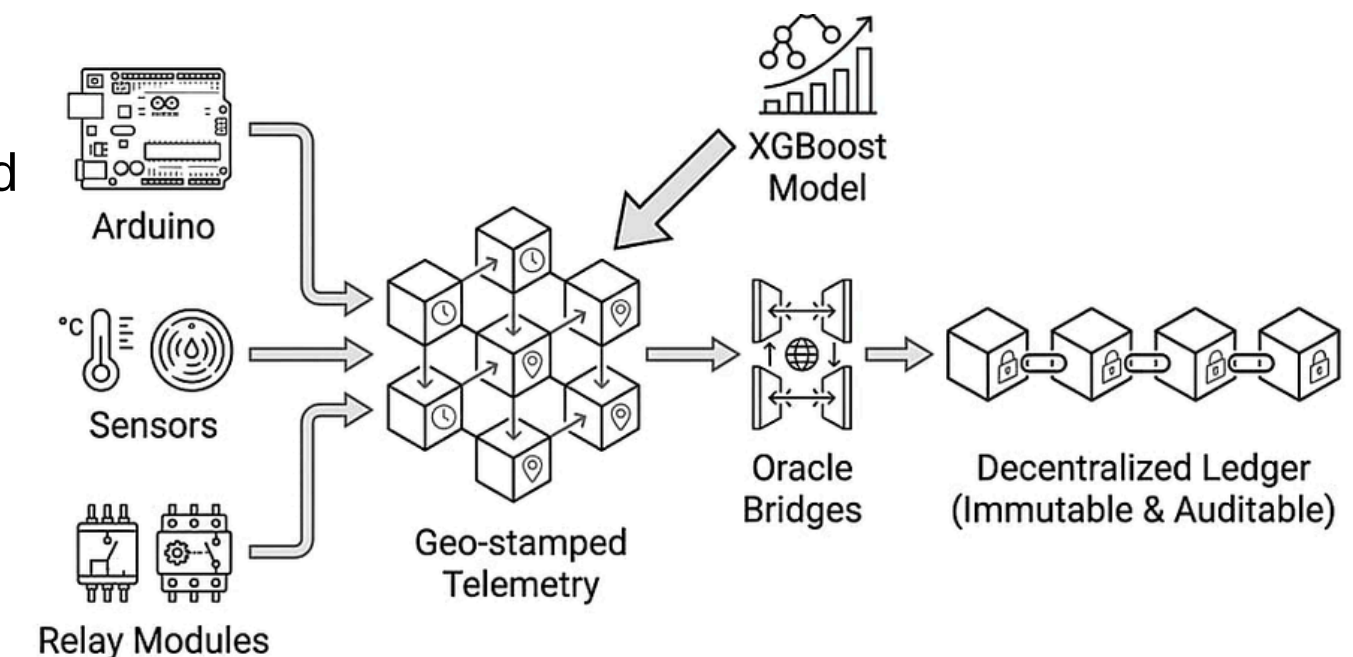
- This foundational layer consists of localized, cost-effective IoT nodes utilizing Arduino, and MQ gas sensors to continuously sample and transmit raw environmental telemetry directly from the community.

LAYER 2: SOFTWARE & AI PREDICTIVE ANALYTICS

- Serving as the user interface, software dashboard translates the raw hardware data into an accessible format. It features dynamic visual metrics, visual analytics and triggers automated health alerts
- Uses **predictive regression AI model** to pattern pollutant dispersion and forecast AQI variations across a 24-hour rolling window.

LAYER 3: BLOCKCHAIN DATA COMMONS

- The secure backend infrastructure that aggregates and logs the anonymous community readings onto a decentralized public ledger, ensuring all environmental data remains transparent, standardized, and strictly tamper-proof



✿ Technology Stack

Hardware & Edge Layer

- Arduino (C / C++ embedded framework)
- Sensor Suite: MQ135, PM2.5 and DHT11

Frontend & User Interface

- Next.js (React) and Tailwind CSS

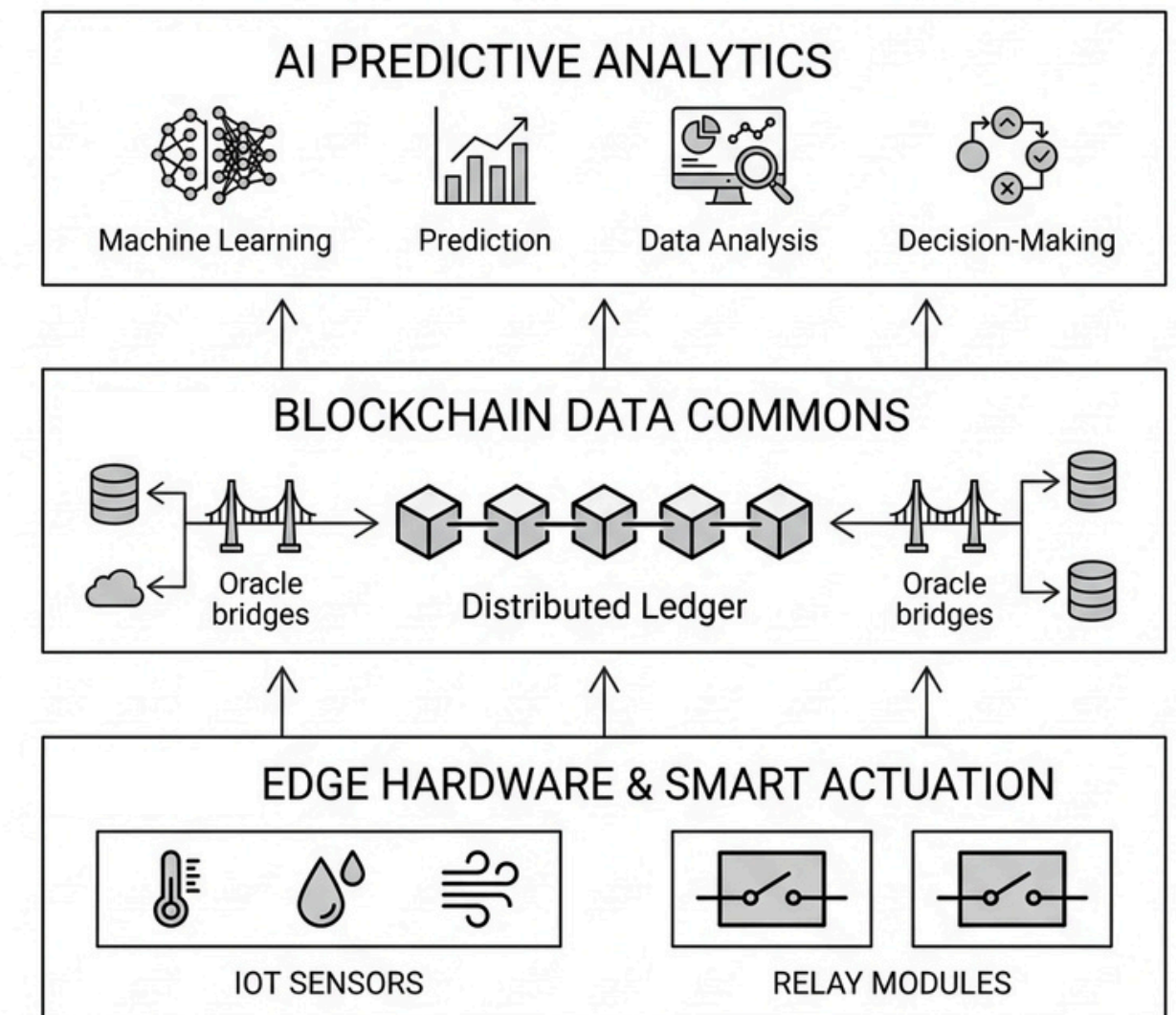
AI & Prediction Engine

- Kriging + Random Forest — interpolates gaps between physical sensors (blind-spot filling).
- Language Models: LLM integration for conversational querying and contextual environmental reasoning
- XGBoost / LightGBM — converts raw satellite AOD into ground-level PM2.5 estimates.

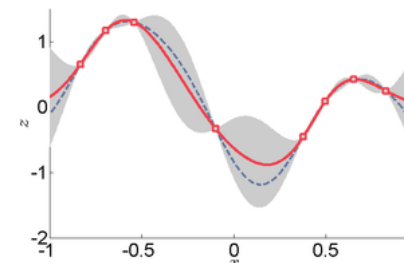
Backend & Data Integrity Layer

- Decentralized Storage & Consensus: Blockchain distributed ledger / Smart Contracts for immutable record logging
- Mappls for live mapping (India-built, used by ISRO & government systems)

A.U.R.A. — Air Utility & Response Architecture



XGB

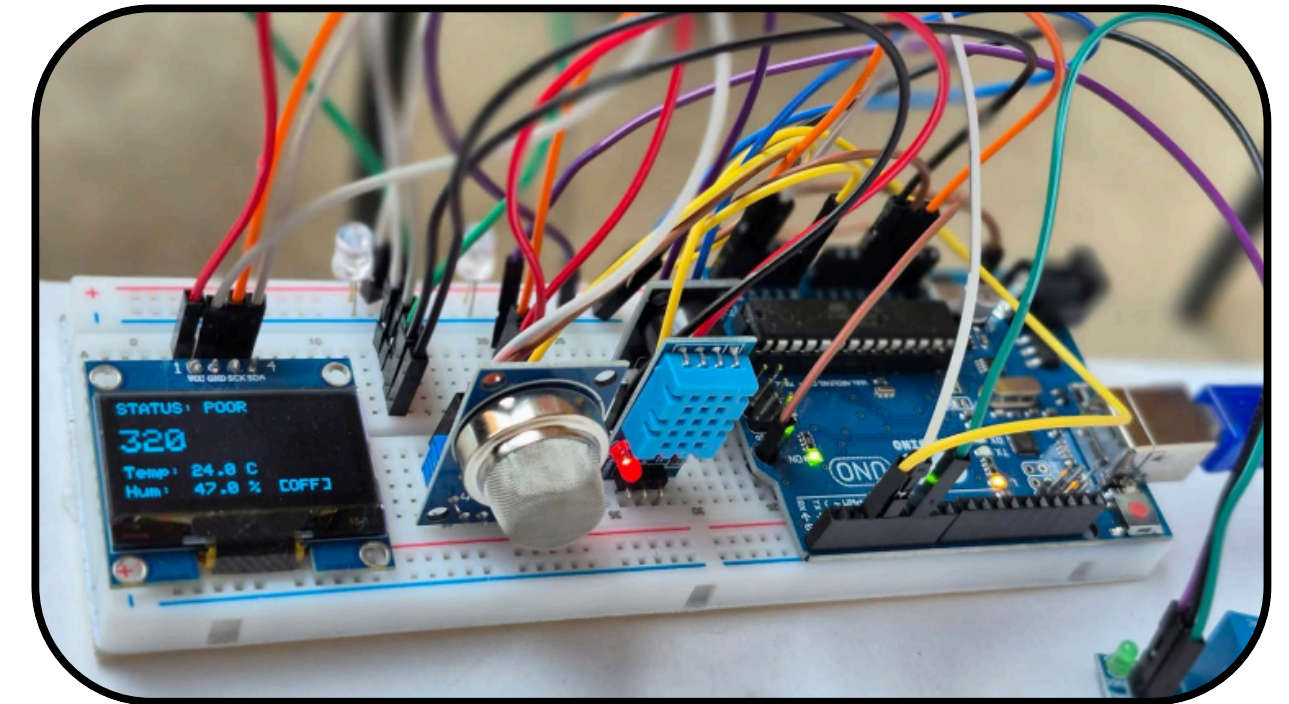


MAPPLS
MapmyIndia



❁ FEASIBILITY

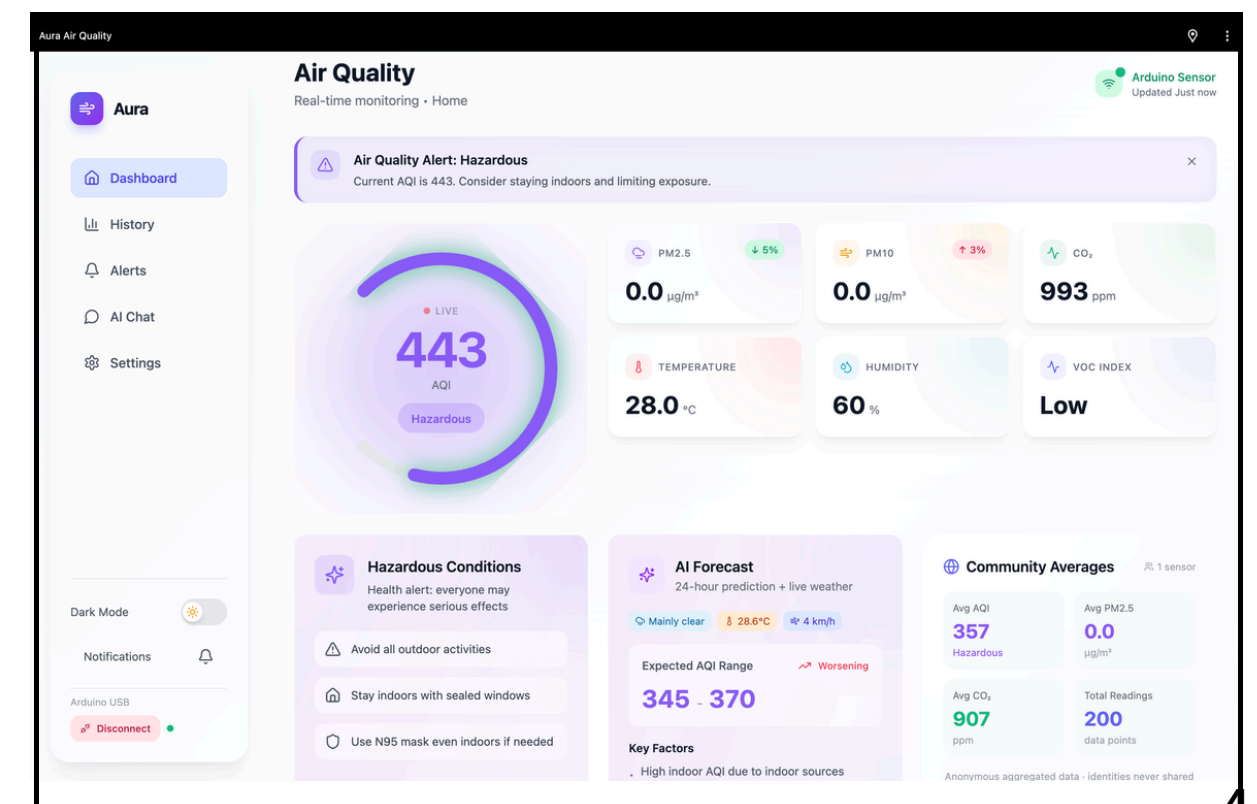
- **Cost-Effective Hardware:** Utilizes affordable, off-the-shelf components (Arduino, MQ sensors), making large-scale community deployment financially viable compared to industrial stations
- **Low Maintenance Operations:** Automated node calibration, real-time wireless transmission, and AI-driven data processing minimize the need for manual intervention or upkeep.
- **Open-Source Adaptability:** The modular software and hardware architecture allows for easy troubleshooting, community contributions, and continuous system iteration.



A.U.R.A. air monitoring device

❁ VISIBILITY

- **Translating the Invisible:** Transforms complex, microscopic environmental data (like PM2.5 and VOCs) into clear, actionable visual metrics on a user-friendly public dashboard.
- **Tamper-Proof Transparency:** Blockchain integration ensures that historical environmental data remains permanently visible to the public and completely resistant to manipulation.
- **Grassroots Awareness:** Decentralized data collection empowers neighborhoods to clearly see their hyper-local air quality, fostering community awareness and advocacy.



A.U.R.A. software's dashboard

✿ Enhancing Public Health and Daily Decision-Making

- Hyper-Local Awareness: Highly localized Air Quality Index (AQI) readings, allowing individuals to make informed daily decisions about outdoor activities, commuting, and exercise.
- Proactive Health Measures: Real-time data access allows vulnerable populations, such as those with respiratory issues, to take immediate precautions during severe pollution spikes.

✿ Fostering Trust and Data Integrity

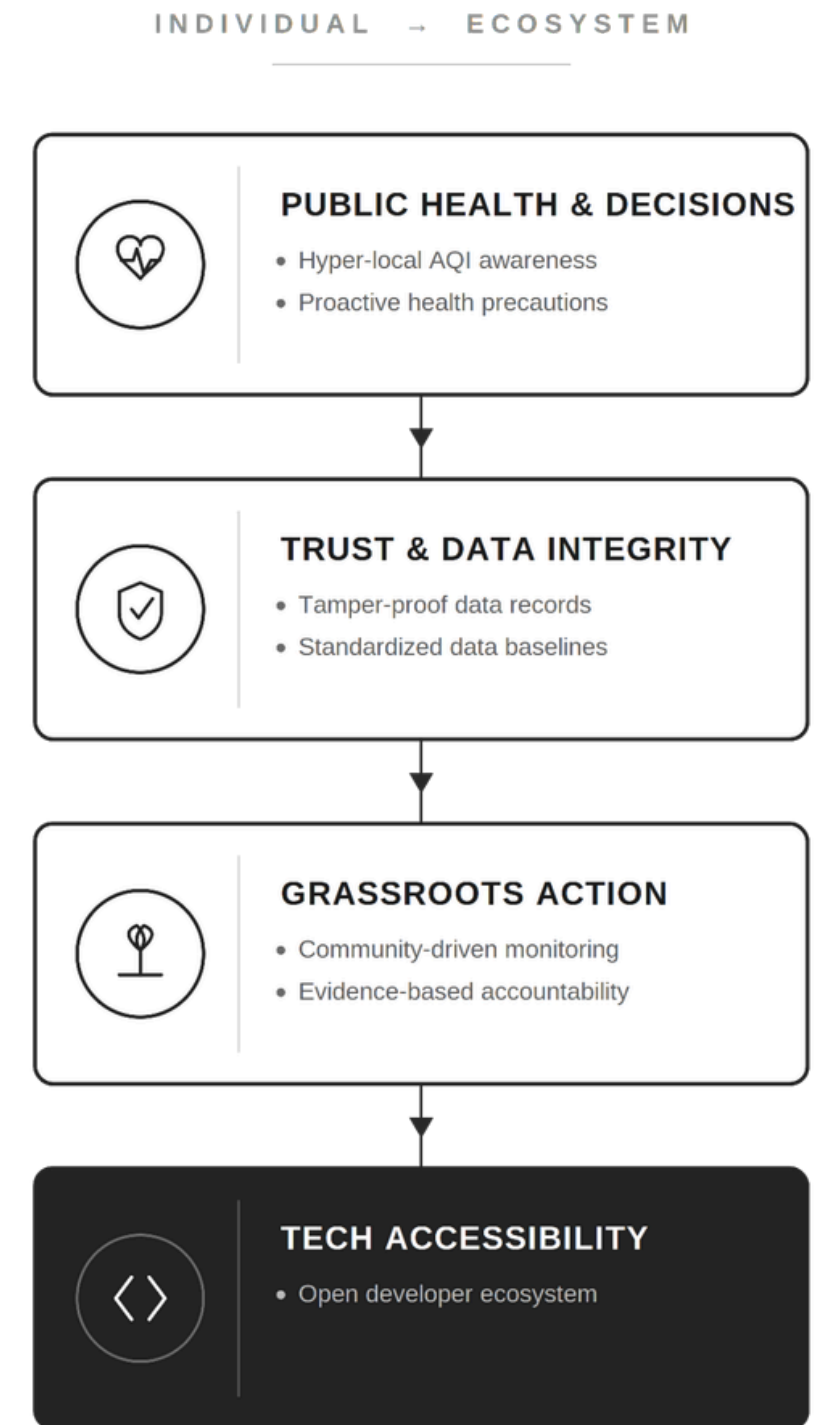
- Tamper-Proof Environmental Records: Utilizing a blockchain ledger ensures that the collected AQI data is completely immutable. This establishes a transparent, undeniable record of environmental conditions that cannot be manipulated by third parties or corporate entities.
- Standardized Baselines: Establishing a reliable, transparent source of truth for air quality allows researchers, civic tech organizations, and health professionals to analyze long-term environmental trends with absolute confidence.

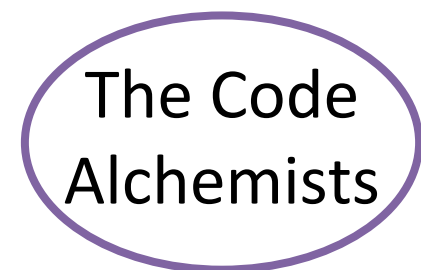
✿ Empowering Grassroots Environmental Action

- Democratized Monitoring: Leveraging cost-effective hardware like Arduino and ESP32 lowers the barrier to entry, enabling everyday citizens, schools, and local communities to actively participate in building the sensor network.
- Policy and Accountability: Communities can leverage the immutable blockchain data to hold local polluters accountable and advocate for stricter emission regulations using hard, verifiable evidence.

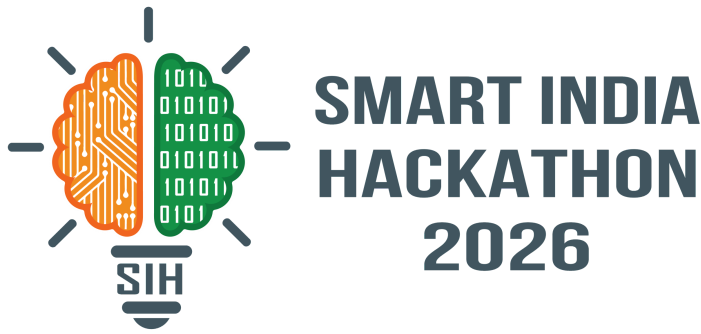
✿ Technological Accessibility

- Open-Source Innovation: Providing an accessible network of standardized readings encourages other developers to build secondary applications on top of the AURA architecture, such as predictive health advisory apps or automated smart-home ventilation systems.





RESEARCH AND REFERENCES



Core Statistics & Sources

- [Lancet Planetary Health, GBD 2019 India Study](#)
- [WHO Global Air Quality Guidelines, 2021](#)
- [CPCB – National Air Quality Monitoring Programme](#)
- [Dalberg / Clean Air Fund, India Report 2026](#)

Data Sources

- [OpenAQ](#)
- [World Air Quality Index \(WAQI\)](#)
- [Copernicus Sentinel-5P \(TROPOMI\)](#)
- [NASA MODIS/VIIRS Aerosol Data](#)

Methodology References

- [Kriging & IDW PM2.5 Mapping – Delhi](#)
- [Random Forest PM2.5 Modeling – NCR](#)
- [RF + XGBoost PM2.5 Estimation – Maharashtra](#)
- [RF/XGBoost/Deep Learning PM2.5 Prediction – MDPI Atmosphere](#)

Tech Stack

- [Mappls Developer Portal](#)

